

Project Proposal

Leveraging Natural Language Processing for Structured Diabetes Data Extraction from Clinical Notes

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Abstract

The growing interest in management of big data, meaningful use of clinical and high quality of care has led to an increase in the demand for biomedical language processing and future insight predict an exponential rise. Free text like clinical notes are messy but rare also the richest and yet most unstructured component of the electronic health record (EHR), they containing valuable information like laboratory values, prescribed medication, disease complications, and important details essential for chronic disease management. This project aims on building from the early initiative by Dr. Lawrence Weed of Problem-Oriented Medical Record (POMR) and the SOAP framework, which argued that medical records should be structured around problems which patient presents upon admission using modern natural language processing (NLP) techniques.

The proposal is a diabetes-focused NLP pipeline that applies tokenization, entity recognition, relation extraction, and normalization to transform free-text notes into structured, interoperable data. By mapping extracted concepts such as HbA1c, insulin regimens, and diabetic complications to standardized vocabularies (LOINC, RxNorm, SNOMED), our system generates timelines of laboratory trends and treatment histories that align with clinical reality. In parallel, hybrid summarization methods integrate extractive and abstract approaches to produce concise patient summaries with provenance links to source text.

This project demonstrates how foundational principles introduced by Weed—structured representation of patient problems—can be extended with large-scale computational methods to address contemporary challenges in diabetes care. Our approach not only reduces the cognitive burden of navigating multi-year notes but also enhances decision support, interoperability, and research utility.

1. Introduction

Diabetes is one of the most data-intensive chronic diseases in modern healthcare with each patient generating a continuous stream of information from lab results such as HbA1c, fasting blood glucose, treatment routines including both orals and injectables, notes on lifestyle interventions, and notes from complications and comorbidities. (Sapra & Bhandari, 2025) This data accumulates with each clinical visit over the years making it cumbersome to track and even find trends in patient history. Electronic health records (EHRs) do capture this patient data in structured tables with prescribed medications and other relevant findings however the majority of the clinically relevant details remain buried as free text. (Ehrenstein et al., 2019) Most of the documented clinical findings by clinicians is in narrative form (“poor glycemic control”, “change increase metformin dosage to 1000mg bd”, “change to SGLT2 due to increase in HB1Ac”) , they are noisy

inconsistent and difficult to process at scale however they have huge importance and rich in meaning.(Kariotis et al., 2022) Physicians use different abbreviations and medical terms to mean the same thing (hypoglycemia ,low blood glucose, low sugar levels),this makes it hard to extract and track longitudinal data reliably. This results in data being scattered across multiple narrative notes and being hard for doctors, researchers and even computers to see timeline quickly .Clinicians have to end up scrolling years of text to reconstruct what happened and this is time consuming and prone errors especially in emergency situations. Clinical decision support systems depend on structured inputs if the notes are not coded high chance system misses it or gives wrong alerts. EHR were meant to make data usable however most of the clinically meaningful details are still trapped in unstructured text, this leaves a gap in patient care and research.

Problem Statement:

Clinical notes are very important in diabetes care and research because they include laboratory result(like HbA1c,glucose,GFR,),medication details(start, stop dosages, insulin records) and complications and comorbidities, however most of this is written in free text which is unstructured .This is a problem because medical vocabulary varies from one physician to another with multiple terms being used to denote the same thing as information is spread across years of text and also computers cant interpret free text accurately. This results in clinicians have to spend time digging through notes by hand ,researchers not being able to use it for large studies, decision support systems ignoring it and valuable data stays hidden Lawrence Weed (the father of the problem-oriented medical record and SOAP notes)wanted medical records to be structured and organized his vision based on sorting information in a way that helps reasoning and decision making but today EHR notes are unstructured again This project aims at bringing Weeds idea into the modern world using NLP and LLMs through extraction of diabetes specific information then linking them to dates and context and then normalize them with standardized codes before summarizing into concise reports.(Patel & Kannampallil, 2015)

Objectives:

1. Extract important details from clinical notes like
 - HbA1c values
 - Medications
 - Complications

Making sure we catch synonyms and abbreviations (A1c vs HbA1c)

2. Link the information extracted together connecting values and events to their context

- “A1c of 8.9% in June 2024”
 - “Changed to SGLT2 and increased metformin dosage from 500mg to 1000mg after that high”
- and handle negations
3. Map extracted terms to medical terminology as to make them consistent ensuring interoperability across HER systems
 4. Create structured outputs that are short and precise supporting clinicians and researchers as well as decision support tools

2. Literature Review / Related Work

Literature Review / Related Work

The application of **natural language processing (NLP)** in biomedicine and healthcare has rapidly expanded, enabling extraction of valuable information from unstructured clinical texts such as electronic health records (EHRs) and supporting clinical decision-making. Diabetes produces a lot of clinical unstructured data which is difficult for computers to analyze ,our project aims to leverage Natural language processing to solve this by turning messy notes into structured data that can be analyzed ,used in clinical decision support systems as well as disease progression models.

Key Studies and Foundations

During the pandemic an NLP application was developed within 10 days at huge scale to analyze clinical text lab reports and even public data from social media, this proved how NLPs can handle real time high volume biomedical text. This success in infectious disease is evident that other chronic diseases like diabetes where timely extraction of labs, treatments and complications is equally important for patients’ outcomes can benefit from this model.(Al-Garadi et al., 2022) .Researchers applied the same method used in diabetes care in :

- finding complications from notes like neuropathy or kidney disease (Saban et al., 2025)
- pulling out HbA1c values from past clinical notes and building timeline of how patient’s diabetes control has changed thus reconstructing glycemic history(Colmenares-Mejia et al., 2025)

- detecting adverse events from drugs like insulin even when not structured in proper fields(Zheng et al., 2022)

MedLEE(Medical Language Extraction and Encoding System) was one of the early systems for medicine inputting free text to produce structured codes from messy doctor notes MetaMap developed by U.S. National Library of Medicine mapped words from UMLS(Unified Medical Language System) giving researchers tools to normalize medical language. Lastly cTAKES was another open-source NLP used for entity extraction from EHRs all these were rule based or dictionary based working well for basic extraction but struggled with context. Modern NLP like BioBERT,ClinicalBERT are deep learning based that learn from patterns in large datasets. This project builds on the early NLP systems ability to extract entities but improves it using deep learnings context awareness which is exactly what we need for diabetes data like HbA1c values,medication,complications,with negation and time.(Alsentzer et al., 2019)

Research Gaps

Despite significant progress, persistent gaps have been identified:

- **Complex Relation Extraction:** Many NLP systems can find individual terms like ‘A1c’(lab test), ‘lantus’(insulin) but have troubles linking them together for example “On June 14 2024 A1c was 9.6%” system needs to connect test to value to unit to date.Our pipeline will be able to handle entity relation extraction to build meaningful A1c and medication histories.
- **Temporal Reasoning(time expressions):** Clinicians often write vague statements like “last years A1c”, ‘started metformin in 2023’, ‘no complications so far’.Many NLP systems don’t normalize this into real dates despite this being very important in progressive diseases like diabetes. This project addresses this by adding temporal normalization (e.g., mapping ‘last year’ to ‘2024-06-14’)
- **Normalization and Coding:** Clinicians use multiple medical vocabularies to denote the same thing and also different brand names of the same drug now NLP has to map all these to standard codes and many systems usually struggle hence reducing interoperability. This project will emphasize on that making data interoperable
- **Generalizability and Evaluation:** Our project aims at making sure our diabetes NLP system won’t only work with one type of dataset but also perform well in other hospitals with different writing styles and EHR systems. Many NLP models are trained on one dataset and when tested elsewhere accuracy drops. Diabetes is everywhere so NLP systems should ideally work across different hospitals and populations. To improve generalizability, we will use rule based and ML

hybrids(rules like NegEx work in any hospital),train with public datasets(i2b2,n2c2,MIMIC)to expose system to different styles and normalize terms to standard vocabularies.

- **Integration Into CDSS:** As most NLP systems stop at extraction and don't integrate into CDS our project addresses this by producing structured interoperable outputs that can be plugged into CDS.

Conclusion

NLP plays a crucial role in diabetes care by enabling the extraction of valuable information from unstructured clinical notes. However, to maximize its impact, persistent challenges such as complex relation extraction, robust normalization, and integration into clinical decision support (CDS) systems must be addressed. Our project aims to contribute to this effort by building upon the foundational ideas of Dr. Lawrence Weed and advances in modern NLP. Specifically, we focus on developing methods for relation extraction and structured representation of diabetes-related information including laboratory values, medications, and complications with the goal of producing interoperable outputs that can support more effective monitoring and decision-making in clinical practice .(Monshi et al., 2020)

3. Methodology

Data Collection

The study will rely on de-identified and publicly available electronic health records (EHRs). The primary dataset will be **MIMIC-IV**, which contains anonymized clinical notes from patients admitted to critical care units. This dataset is widely used in biomedical informatics research and ensures compliance with privacy regulations.

In addition, if the dataset doesn't have enough diabetes cases we can use synthetic data generated using tools such as *Synthea* to supplement the dataset, ensuring balanced representation across diabetes-related cases while protecting patient privacy.

MIMIC-IV dataset: <https://physionet.org/content/mimiciv/>

Synthea synthetic dataset generator: <https://synthetichealth.github.io/synthea/>

Data Preprocessing

Before model training, the raw clinical text will be prepared to ensure quality and consistency:

- **Cleaning:** Remove unnecessary characters, anonymize any identifiers, and fix formatting issues.
- **Normalization:** Map terms such as “blood sugar” and “glucose” to standardized codes using vocabularies like **UMLS, SNOMED CT, LOINC, and RxNorm**.
- **Segmentation:** Split notes into meaningful units (sentences, sections).

Dataset Validation and Relevance

Our project will use data from MIMIC-IV and Synthea to train and evaluate our model. This dataset is extensively validated and used in hundreds peer reviewed healthcare AI studies which confirms its reliability for biomedical NLP research providing both structured and unstructured data. MIMIC-IV will provide validated real world EHR data while Synthea contributes synthetic, privacy-safe variety together forming robust foundation for training and evaluating the diabetes focused NLP pipeline. Their combination supports both technical accuracy (through real data) and ethical compliance (through synthetic data), ensuring that the system can effectively extract, normalize, and summarize diabetes-related information from clinical notes across diverse healthcare environments.

Model Development Pipeline

The project will implement a structured NLP pipeline with the following steps:

1. **Tokenization** – Break clinical text into words or meaningful chunks using *spaCy* or *scispaCy*.
2. **Named Entity Recognition (NER)** – Identify key diabetes-related concepts such as lab values (HbA1c, glucose), medications (insulin, metformin), and complications (neuropathy, retinopathy). This extracts key diabetes information from messy clinical notes
3. **Relation Extraction** – Link entities together. Instead of random facts we will be able to know who the facts belong to and how it connects
4. **Normalization** – Map extracted entities and relations to standardized medical codes to ensure interoperability across systems.
5. **Summarization** – Generate concise summaries of patient diabetes history, focusing on lab trends, medication changes, and complications. Both rule-based and machine learning approaches will be considered.

Model Training and Evaluation

We will teach the model to do NER and relation extraction correctly by:

- Using labeled data annotated by humans or existing corpora (by borrowing some existing labeled data (i2b2, n2c2), then add a small set of our own manual annotations for diabetes-specific relations (like HbA1c trends) this combination will give us enough training material for the model.)
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- Fine tune models like ClinicalBERT
- Split into training/validation/test sets.
- Evaluate with precision,recall,F1 score

This proves our system works and is reliable

Workflow Summary

1. Collect and preprocess raw EHR notes.
2. Extract diabetes-related entities (labs, medications, complications).
3. Link related entities through relation extraction.
4. Normalize extracted concepts into standard vocabularies.
5. Summarize outputs for integration into clinical decision support.

Tools: Mention spaCy, MedSpaCy, scispaCy, regex, UMLS APIs.

Workflow Diagram: Insert figure here.

7. References

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